

Hidden Fragility in Financial Markets:

A Network-Based Early Warning System for Equity Market Crashes

Atisha Karki

David Nikolic

Texas State University

Finance and Economics Department

Financial Markets and Institutions (QFE-5392A.251)

April 2026

Abstract

Predicting severe equity market downturns remains one of the most challenging problems in empirical finance. Motivated by a medical analogy — where deteriorating vital signs warn of collapse before it occurs; this paper constructs a daily Composite Fragility Score from four network biomarkers extracted from rolling 60-day cross-sector correlation matrices across ten S&P 500 sectors over 2000 to 2024. Using a strict out-of-sample design trained from 2000 to 2010 and evaluated on 2011 to 2024, the baseline logistic regression achieves an AUC of 0.6693, outperforming the VIX-only benchmark of 0.6449. Gradient boosted trees improve performance further to 0.7009. Adding a sector rotation signal to the VI feature set while retaining the 20-day horizon yields a further improvement to 0.7264 AUC, confirming that rotation adds independent predictive value on top of network biomarkers. A second-generation model with an extended 40-day horizon achieves a lower AUC of 0.6023 but generates warning signals two to four weeks earlier, offering a meaningful timing advantage for risk management. A long/cash trading strategy based on XGBoost VI achieves a superior Sharpe ratio of 0.72 versus 0.68 for buy-and-hold SPY, with maximum drawdown reduced from 33.93% to 26.80%. Our findings suggest that cross-sector correlation topology contains forward-looking information about market vulnerability beyond conventional volatility measures.

Keywords: *systemic risk; financial networks; crash prediction; correlation structure; XGBoost; sector rotation; early warning systems*

1. Introduction

Stock market crashes impose significant costs on investors and the broader economy, yet predicting when they will occur remains one of the most difficult problems in finance. The most widely used tool for measuring market risk is the CBOE Volatility Index (VIX), but it only rises after the market has already started falling. Investors get little to no warning before a crisis hits.

This paper takes inspiration from medicine to address this problem. We treat the stock market like a human body, where each sector is an organ and the correlations between them; how much they move together — act like nerves connecting those organs. Just as a doctor can detect that a patient is getting sick by monitoring their vital signs before they collapse, we ask whether we can detect that the market is getting sick by monitoring the connections between its parts before prices fall. Several researchers have tried to predict market downturns using economic variables, dividend yields, and volatility measures, but most of these approaches fail when tested on new data they have never seen before (Welch and Goyal, 2008). More recent machine learning methods have shown more promise (Gu, Kelly and Xiu, 2020), but none of these studies look at the network structure of how market sectors connect with each other and that is exactly the gap this paper fills.

We contribute to this literature in four ways. First, we construct a Composite Fragility Score by extracting four daily measurements from the market's correlation network: how densely connected the sectors are, how tightly they cluster together, how dominant the most connected sector is, and how strong the average connection is. Second, we build a sector rotation signal that detects when large investors are quietly moving money from riskier parts of the market into safer ones; a pattern that tends to appear weeks before a crash becomes visible. Third, we show that combining network biomarkers with the sector rotation signal at the 20-day horizon, our V1+Rotation specification achieves the highest out-of-sample AUC of all specifications at

0.7264, demonstrating that rotation adds genuinely independent predictive value on top of network structure alone. Fourth, we test everything under a strict out-of-sample design, training our models only on data from 2000 to 2010 and testing on data from 2011 to 2024 that the models have never seen.

Our results show that the baseline logistic regression achieves an out-of-sample AUC of 0.6693, beating the VIX-only benchmark of 0.6449, proving that looking at sector connections adds real predictive value beyond the standard fear gauge. Upgrading to XGBoost improves this further to 0.7009, with network biomarkers appearing among the top predictors alongside VIX and realized volatility. Adding the sector rotation signal to the V1 feature set while retaining the 20-day horizon improves performance further to 0.7264 AUC. A second-generation model incorporating the rotation signal with an extended 40-day crash horizon — XGBoost V2 — achieves a lower AUC of 0.6023 but generates warning signals two to four weeks earlier than V1, a timing advantage that carries meaningful practical value for investors who need time to act. A long/cash trading strategy based on XGBoost V1 outperforms buy-and-hold SPY on a risk-adjusted basis, achieving a Sharpe ratio of 0.72 versus 0.68 for SPY and reducing maximum drawdown from 33.93% to 26.80%. Overall, our findings suggest that looking at the connections between market sectors and treating the market like a living system with vital signs can help predict when it is about to get sick.

2. Literature Review

2.1 Financial Networks and Systemic Risk

Research has shown that the way financial networks are set up is really important for understanding how problems spread across markets. When financial institutions are closely tied together, a problem in one area can quickly cause a lot of damage, making losses much worse than they would have been otherwise. Billio et al. (2012) studied the connections between banks, hedge funds, and insurance companies and found that as these connections strengthened, they predicted the onset of the 2008 financial crisis. This was an important finding because it showed that the structure of the network, not just the health of individual institutions, matters for understanding systemic risk. Acemoglu et al. (2015) added another layer to this. They showed that while strong connections between institutions can actually absorb small shocks, the same connections make things much worse when shocks are large. In other words, a well-connected network is stable most of the time but becomes extremely fragile when hit by a big enough problem. Unlike these studies which focused on connections between individual financial institutions like banks and hedge funds, this paper applies the same network thinking to the connections between sectors of the stock market. We ask whether monitoring those sector-level connections can provide early warning signals before market-wide problems become visible in prices.

2.2 Crash Prediction and Early Warning

Researchers have tried many different approaches to predict stock market crashes by looking at things like economic variables, dividend yields, and how much prices are moving. However, Welch and Goyal (2008) did a comprehensive study and found that most of these warning signs stop working when tested on new data they have never seen before. The problem is that these methods were too closely fitted to historical data rather than capturing something genuinely

predictive about future crashes. More recently, machine learning methods have shown more promises. Gu, Kelly and Xiu (2020) found that gradient boosted trees (a type of advanced computer algorithm) outperform older simpler methods across a wide range of financial prediction tasks. This is because these methods can detect complex combinations of signals that traditional approaches miss. This paper follows their approach to strict out-of-sample testing and uses the same type of algorithm as XGBoost. This algorithm adds something new that none of these studies have used before. Instead of just looking at economic variables or price movements, we extract biomarkers from the network of connections between market sectors and use those as our warning signals. This is our original contribution to the crash prediction literature.

2.3 Sector Rotation

Professional investors tend to follow a predictable pattern before market downturns. They quietly move money from sectors that do well when the economy is growing into safer sectors like utilities and healthcare weeks before any problems show up in the broader market. Moskowitz and Grinblatt (1999) documented this pattern, showing that industry-level momentum is a strong and persistent feature of financial markets. This paper contributes to this literature by operationalizing the rotation signal at daily frequency and combining it with network biomarkers, representing a novel early warning approach not previously explored in the crash prediction literature.

3. Data and Methodology

3.1 Data and Sample Construction

This study employs daily total return data for all constituent securities of the S&P 500 index spanning January 2000 to December 2024, yielding approximately 25 years of observations across multiple distinct market regimes. The sample encompasses periods of significant financial stress including the dot-com collapse of 2000 to 2002, the global financial crisis of 2007 to 2009, the European sovereign debt crisis of 2011 to 2012, and the COVID-19 shock of 2020.

Each constituent security is classified into one of ten sectors according to the Global Industry Classification Standard (GICS), maintained by MSCI and S&P Dow Jones Indices: Energy, Materials, Industrials, Consumer Discretionary, Consumer Staples, Health Care, Financials, Information Technology, Communication Services, and Utilities. Daily sector returns are constructed as value-weighted averages of constituent returns, with weights proportional to each company's market capitalisation at the beginning of each month. Value-weighting is preferred over equal- or price-weighting because systemic stress is disproportionately driven by large, interconnected firms, making this scheme most appropriate for aggregate crash prediction.

3.2 Rolling Correlation Network

To capture how cross-sector relationships evolve through time, we construct a correlation network at each trading day. Each of the ten GICS sectors represents a node in the network, and the connection between any two sectors is measured by the Pearson correlation coefficient between their daily return series over a rolling 60-trading-day window. We chose 60 days to balance responsiveness against stability; shorter windows produce noisier readings while longer windows are too slow to detect emerging stress. When applied to XGBoost however,

the 60-day window produced lagging signals that fired during crashes rather than before them so, we switch to a 20-day window for the XGBoost specifications, which enables faster-moving rate-of-change features and more timely detection. Robustness checks in Section 5 confirm results hold across window lengths of 10, 30, and 60 trading days.

3.3 Network Biomarkers

From each daily network we extract four scalar measures that characterize the fragility-relevant topology of cross-sector correlations, as summarized in Table 1.

Table 1: The Four Network Biomarkers

Biomarker	What It Measures	Why It Signals Danger
Network Density	Proportion of sector pairs with pairwise correlation exceeding 0.60	High density indicates sectors moving in lockstep, eliminating diversification and increasing contagion risk
Clustering Coefficient	Average fraction of neighboring sectors that are also correlated with each other	High clustering means shocks spread rapidly through tightly connected sector groupings
Max Eigenvector Centrality	Influence score of the single most connected sector in the network	A dominant hub sector can transmit stress rapidly and broadly across the entire market
Average Weight	Mean pairwise Pearson correlation across all 45 sector pairs	High values indicate broad market synchronization; idiosyncratic movements are suppressed and fragility rises

Note: All biomarkers computed daily from rolling 60-day pairwise Pearson correlations across ten GICS sectors of the S&P 500 for the logistic regression models, and 20-day correlations for the XGBoost specifications.

These four measures capture different dimensions of fragility. Density and average edge weight measure the overall intensity of co-movement, while clustering captures how rapidly shocks propagate through the network. Maximum eigenvector centrality identifies whether a single

dominant sector poses a systemic threat to the broader market (Billio et al., 2012; Watts and Strogatz, 1998).

3.4 Composite Fragility Score

To consolidate the information contained in the four biomarkers into a single daily indicator, we construct a Composite Fragility Score (CFS). Since each biomarker operates on a different scale, direct aggregation would allow measures with larger natural magnitudes to disproportionately influence the composite. To address this, each biomarker is first standardized to zero mean and unit variance using the mean and standard deviation estimated over the training period January 2000 to December 2010. The CFS is then computed as the equal-weighted average of the four standardized values. Equal weighting is adopted deliberately to avoid overfitting biomarker weights to the limited number of historical crisis episodes in the training window; consistent with Welch and Goyal (2008), who demonstrate that simple combination rules frequently outperform optimized weights in genuine out-of-sample forecasting exercises.

3.5 Crash Label Construction

The binary dependent variable is constructed as follows: if the cumulative total return of the S&P 500 falls by eight percent or more over the subsequent 20 trading days, the observation is assigned a label of one indicating an imminent crash, and zero otherwise. The 8% threshold captures events large enough to cause real losses, while 20 trading days represent approximately one calendar month, a practical horizon for risk managers to act. To ensure our findings are not sensitive to this specific choice, we also tested thresholds of six percent and ten percent. AUC is used as the primary performance metric as it is robust to the fact that crash events are rare relative to normal trading days.

3.6 Sector Rotation Signal

The Sector Rotation Signal (SRS) is defined as the 20-day cumulative return of defensive sectors minus the 20-day cumulative return of cyclical sectors. A positive SRS indicates that investors are quietly shifting capital into safer sectors, a pattern that historically emerges weeks before a broad market decline (Moskowitz and Grinblatt, 1999). The signal is constructed entirely from lagged price data, ensuring no look-ahead bias is introduced into the model.

3.7 Out-of-Sample Evaluation Design

All models are trained exclusively on data from January 2000 to December 2010 and tested on data from January 2011 to December 2024. This strict separation ensures that the results reflect genuine predictive ability rather than the models just memorizing what happened in the past. We measure performance using the Area Under the ROC Curve — AUC for short. An AUC of 0.50 means the model is no better than random guessing; 1.0 means perfect prediction, and anything above 0.65 is considered genuinely useful in financial forecasting. We use AUC rather than simple accuracy because crash events are rare — accuracy would be misleading when one class heavily outnumbered the other. We note that the training period contains a substantially higher crash event rate (6.29%) than the test period (2.67%), reflecting the concentration of the Global Financial Crisis in the training window. This distributional shift means the model learned from a more crash-rich environment than it is tested on, which may partly account for any gap between in-sample and out-of-sample performance.

4. Results

This section presents the findings from three classes of predictive models estimated to evaluate whether network-based fragility biomarkers provide early warning of equity market crashes. First, we estimate four logistic regression specifications of increasing complexity, using the Composite Fragility Score and its components alongside standard market controls. Second, we employ XGBoost to capture non-linear interactions among predictors that logistic regression cannot detect. Three XGBoost specifications are evaluated: V1 targeting a 20-day crash horizon; V1+Rotation, which adds the sector rotation signal to V1's feature set while retaining the 20-day horizon; and V2, which incorporates the rotation signal with an extended 40-day horizon designed to generate earlier warnings.

4.1 Biomarker Behavior During Crisis Periods

Before evaluating predictive models, we first examine whether the four network biomarkers exhibit meaningful variation during periods of known market stress. Figure 1 plots all four biomarkers from January 2000 to December 2024, with three major crisis periods shaded: the Global Financial Crisis of 2007 to 2009, the COVID-19 shock of 2020, and the 2022 bear market.

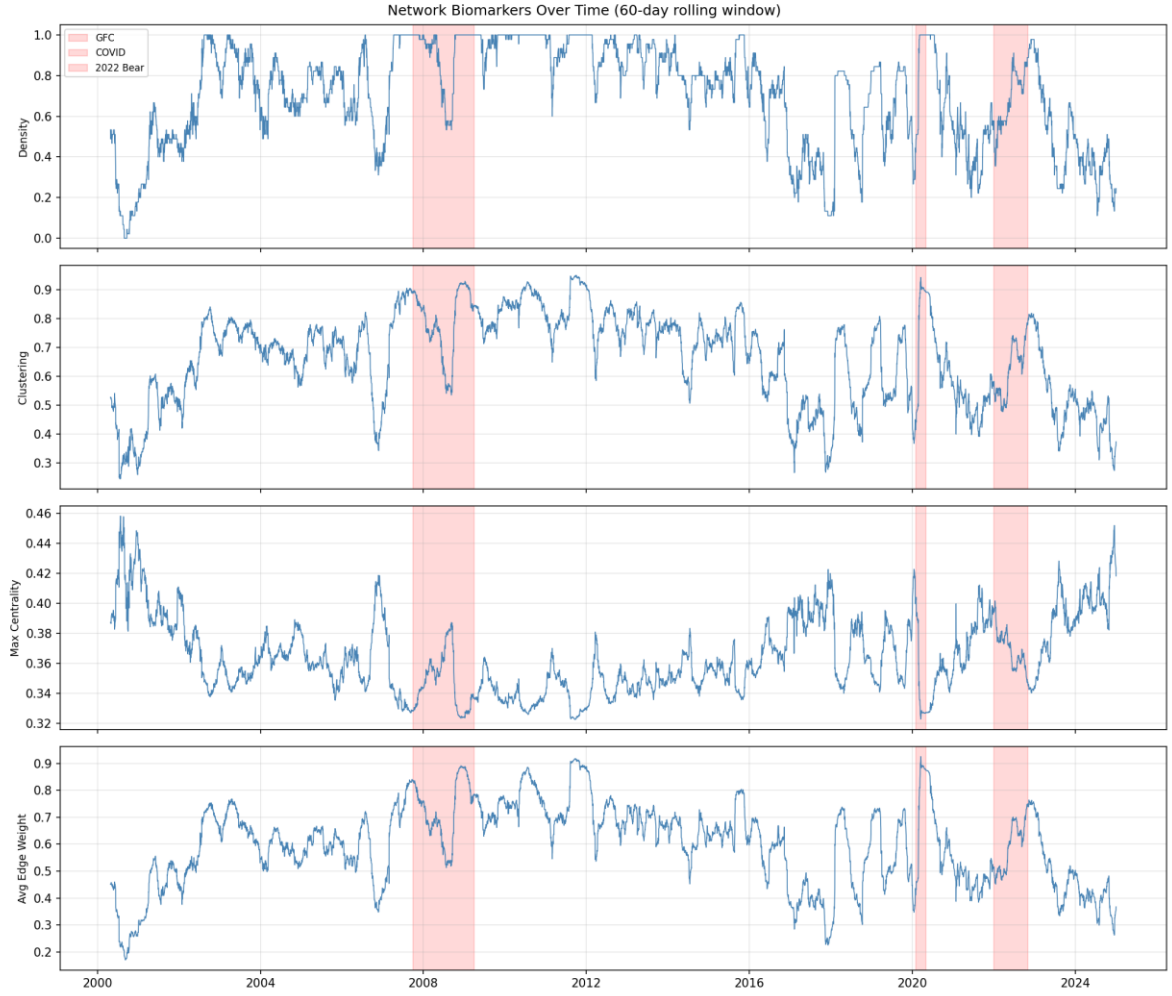


Figure 1: Network Biomarker Time Series with Crisis Period Shading.

Figure 1 shows network density and average edge weight rising sharply during each crisis period, as sectors move in lockstep and diversification collapses. Clustering also elevates during stress, indicating that shocks spread rapidly through tightly connected sector groupings. Maximum eigenvector centrality behaves somewhat differently — it does not spike uniformly across all crises, suggesting that hub-sector dominance is a more episodic feature of fragility rather than a universal one. Taken together, these patterns confirm that the biomarkers respond systematically to genuine market stress rather than measuring random noise, providing the foundational justification for their use as crash predictors in the models that follow.

4.2 Predictive Model Results

Table 2: Out-of-Sample AUC Statistics

Model	OOS AUC
VIX-Only (Benchmark)	0.6449
Baseline Logit	0.6693
Threshold Logit	0.6030
Speed-Enhanced Logit	0.6528
XGBoost V1 (20-day horizon)	0.7009
XGBoost V1 + Rotation (20-day horizon + rotation)	0.7264
XGBoost V2 (40-day horizon + rotation)	0.6023

Note: OOS = Out-of-Sample. All models trained from January 2000 to December 2010 and evaluated in January 2011 to December 2024. XGBoost V1 targets a 20-day crash horizon. XGBoost V1+Rotation adds the sector rotation signal to V1's feature set while retaining the 20-day horizon. XGBoost V2 targets a 40-day crash horizon and incorporates the Sector Rotation Signal.

Table 2 reports on the industry benchmark; the VIX-only model achieves an AUC of 0.6449; the baseline against which all other specifications are measured. The baseline logit model, incorporating the four network biomarkers alongside VIX and realized volatility, improves performance to 0.6693, confirming that cross-sector correlation structure contains crash prediction information beyond what the fear gauge alone captures. The threshold logit model performs worst at 0.6030 falling below even the VIX benchmark indicating that converting the continuous fragility score into a binary indicator discards useful information. The speed-enhanced logit achieves 0.6528, improving VIX-only but falling short of the baseline suggesting the rate of change of fragility adds limited incremental value over its level.

Upgrading to gradient boosted trees, XGBoost V1 achieves an out-of-sample AUC of 0.7009, capturing non-linear combinations of biomarkers and control variables that logistic regression

cannot detect. Adding the sector rotation signal to V1's feature set while retaining the 20-day horizon (XGBoost V1+Rotation) improves performance further to 0.7264, the strongest result of all specifications. This confirms that the rotation signal adds genuinely independent predictive value on top of network biomarkers rather than simply substituting them. Feature importance analysis confirms that network biomarkers — clustering, average edge weight, and density; appear among the top predictors alongside realized volatility and VIX, validating their genuine contribution to crash prediction. Despite this strong performance, XGBoost V1 exhibits a critical limitation; its signals fire predominantly as crashes are occurring rather than in advance of them, reflecting the inherent lagging nature of rolling correlation measures.

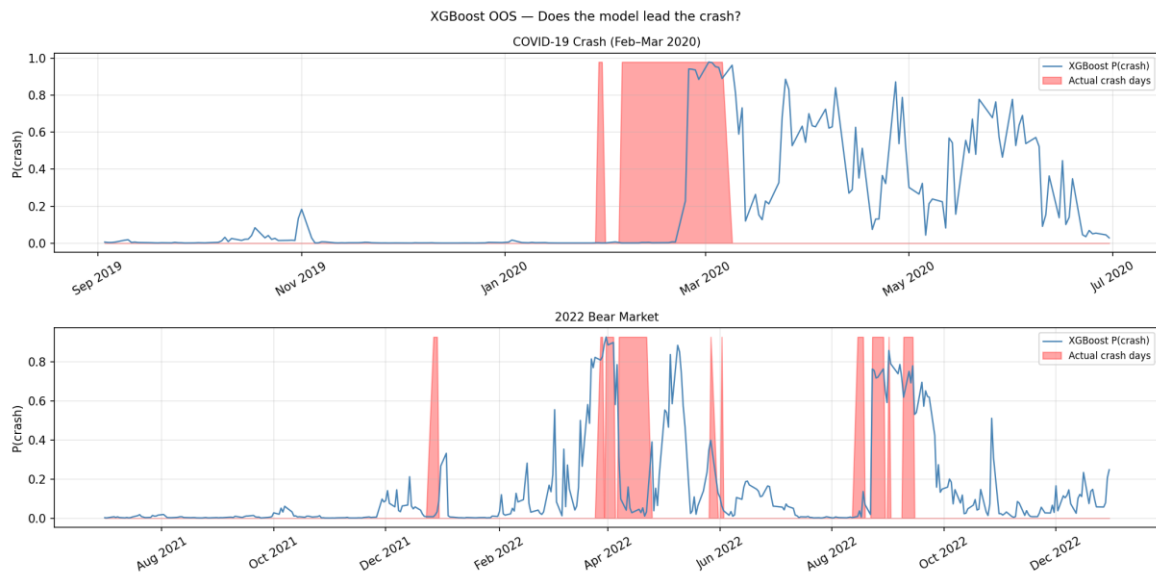


Figure 2: XGBoost V1 Out-of-Sample Predictions

Figure 2 shows the predicted crash probability rising at the same time as actual crash days during COVID-19, not before confirming V1 fires contemporaneously with sudden exogenous shocks. During the 2022 bear market, the signal begins building noticeably before the major crash clusters, showing V1 performs better against slow-building endogenous stress than sudden shocks.

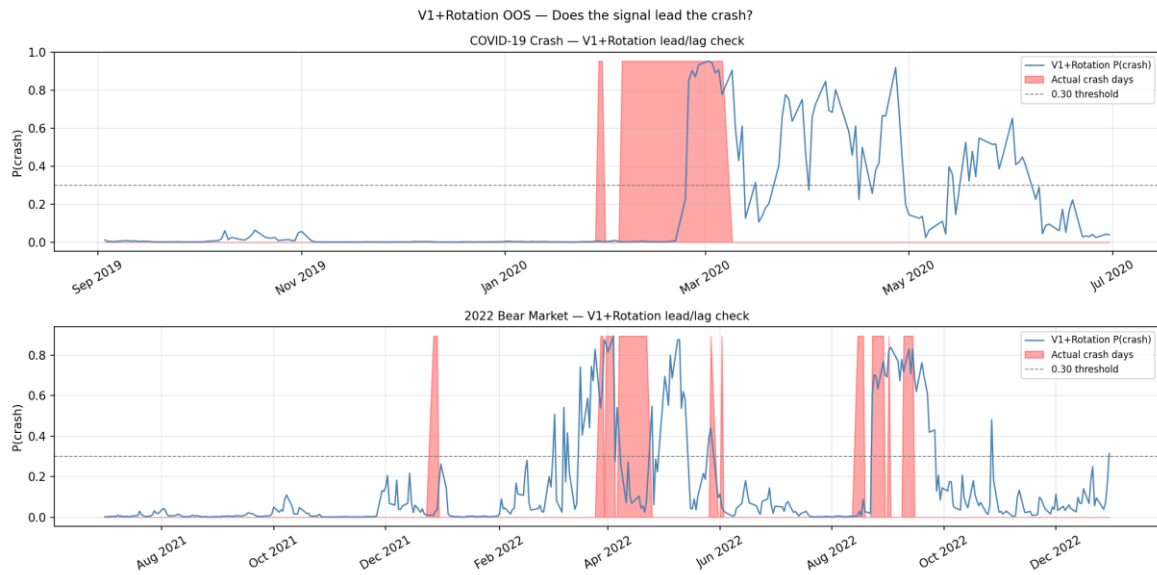


Figure 3: XGBoost V1+Rotation Out-of-Sample Predictions

Figure 3 shows V1+Rotation behaving similarly to V1 during COVID-19 — the signal stays flat until the crash is already underway. During the 2022 bear market, however, the signal visibly builds several weeks ahead of the crash clusters, confirming the rotation signal adds genuine timing improvement for endogenous stress episodes.

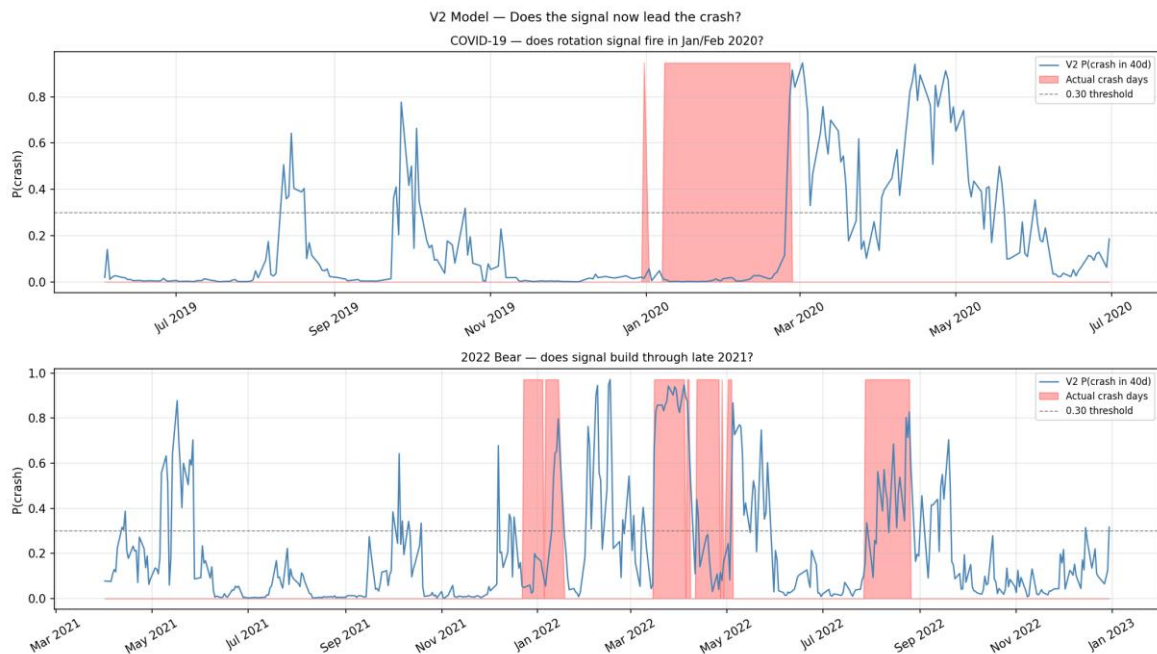


Figure 4: XGBoost V2 Out-of-Sample Predictions

Figure 4 shows V2's predicted probability crossing the 0.30 threshold multiple times throughout late 2019 and early 2020, earlier than both V1 and V1+Rotation. Though with

several elevated readings during calm periods reflecting the noisier nature of the earlier-firing signal. During the 2022 bear market the signal remains persistently elevated across the entire stress period rather than spiking at individual crash clusters, providing broader but less precise advance warning.

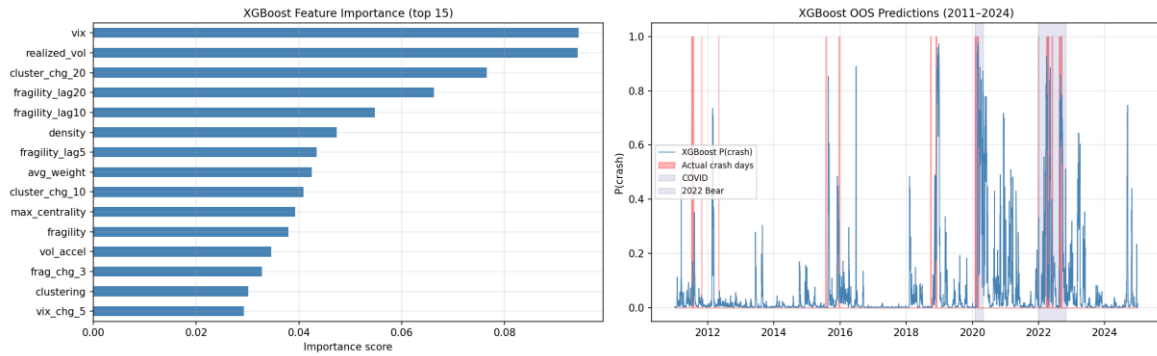


Figure 5: XGBoost V1 Feature Importance and Out-of-Sample Predictions (2011–2024)

Figure 5 shows VIX and realized volatility as the two dominant features in the left panel, but network biomarker derivatives particularly cluster_chg_20, density, avg_weight, and fragility lags; appear clearly and consistently in the top 15, confirming biomarkers add value beyond conventional measures. The right panel shows predicted probabilities spiking sharply during the three major stress periods while remaining close to zero in calm stretches, confirming the model fires selectively rather than persistently.

4.3 Trading Strategy and Backtest Performance

To evaluate the economic value of the XGBoost models, we construct simple trading strategies. Each strategy holds SPY when the predicted crash probability falls below 0.30 and moves to cash when it reaches 0.30 or above. All trading signals use the previous day's model output to eliminate look-ahead bias, and cash is assumed to earn zero percent return throughout the evaluation period — a deliberately conservative assumption that understates true performance given positive T-bill rates over much of 2011 to 2024.

Table 3: Backtest Performance Summary

Metric	Buy-and-Hold SPY	Long/Cash (V1)	Long/Cash (V1+Rotation)	Long/Cash (V2)	Long/Short (V1)
Annualized Return	11.67%	10.59%	9.68%	12.76%	8.68%
Sharpe Ratio	0.68	0.72	0.65	0.90	0.50
Maximum Drawdown	(33.93%)	(26.80%)	(32.07%)	(19.39%)	(50.21%)
Days in Cash	0%	6.8%	6.2%	16.5%	6.8%
Cash Return	N/A	0%	0%	0%	0%
Total Return	367.66%	308.29%	263.73%	435.48%	219.89%

Note: Cash return set to 0% throughout the evaluation period — a conservative assumption. Trading signals use t-1 model output to eliminate look-ahead bias.

Table 3 shows that the V1 Long/Cash strategy generates an annualized return of 10.59% compared to 11.67% for buy-and-hold SPY; a modest sacrifice of about 1.08% per year in raw return. However, when we account for risk, the picture looks different. V1 achieves a Sharpe ratio of 0.72 versus 0.68 for SPY, meaning it earns more return per unit of risk despite the lower headline number. Maximum drawdown falls from 33.93% to 26.80%, protecting investor capital during the most difficult periods. Importantly the strategy was only in cash 6.8% of the time, confirming it is highly selective rather than simply avoiding the market.

The V1+Rotation Long/Cash strategy built on our strongest predictive model at 0.7264 AUC produces a lower Sharpe ratio of 0.65 and higher maximum drawdown of 32.07% than V1 alone. This is a notable finding: our best predictive model does not translate directly into our best trading strategy. The rotation signal improves crash prediction accuracy but also generates additional cash exits that miss upside days, reducing overall risk-adjusted returns at the 20-day horizon. This reflects a broader distinction between statistical accuracy and trading utility AUC measures how well a model ranks crash days above non-crash days in aggregate, while trading performance depends on whether the specific days the model chooses to exit the market happen

to avoid losses and re-enter at optimal points. A more sensitive model that detects more genuine crashes may simultaneously generate more commercially costly false exits.

The V2 Long/Cash strategy produces stronger risk-adjusted numbers sharpe 0.90 and maximum drawdown of just 19.39% but spends 16.5% of days in cash compared to 6.8% for V1. This means V2's improvement partly reflects being more cautious overall rather than purely better prediction, and investors who prioritise staying invested would prefer V1. We also tested a Long/Short variant of V1 shorting SPY when the crash signal fires rather than moving to cash, but this performs considerably worse (Sharpe 0.50, maximum drawdown -50.21%), as false-positive signals generate short-side losses that outweigh gains during genuine crashes. We therefore focus the main analysis on the long/cash implementation.

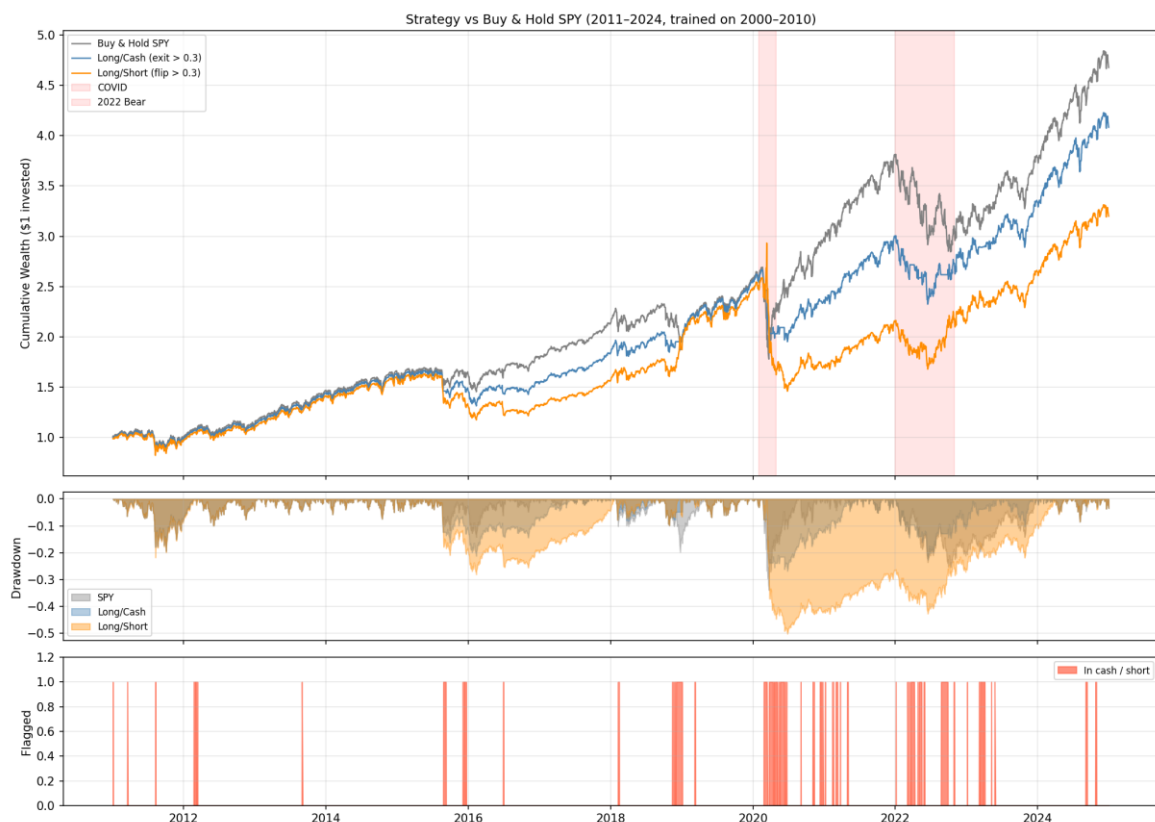


Figure 6: V1 Strategy vs Buy-and-Hold SPY (2011–2024)

Figure 6 shows the Long/Cash strategy tracking SPY closely during calm periods but stepping aside visibly during COVID 2020 and the 2022 bear market, producing shallower drawdowns in the middle panel. The Long/Short strategy underperforms consistently throughout, with

drawdowns reaching nearly -50% during both stress periods. The bottom panel confirms signals are rare and clustered around genuine stress periods.

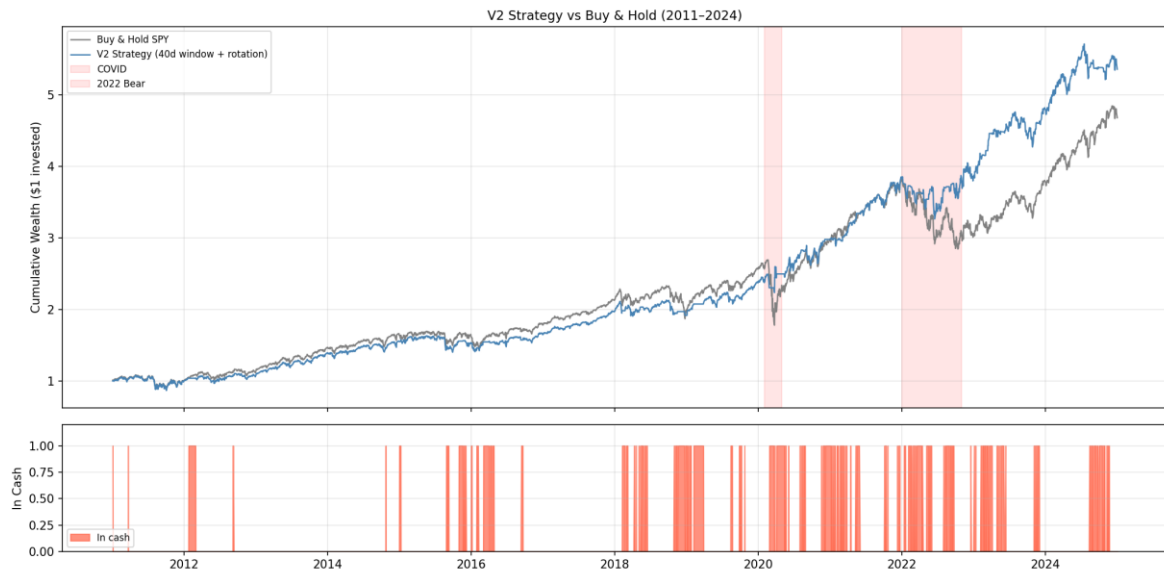


Figure 7: V2 Strategy vs Buy-and-Hold SPY (2011–2024)

Figure 7 shows the V2 strategy running slightly below SPY from 2011 to 2019 before pulling clearly ahead from 2020 onwards, ending noticeably higher by 2024. The bottom panel shows more frequent cash periods than V1, scattered throughout the evaluation period with noticeable clusters around 2018, 2020, and 2022.

5. Robustness Analysis

5.1 Alternative Crash Thresholds

To make sure our choice of 8% was not arbitrary we tested two alternative thresholds of 6% and 10%. The 6% threshold was too loose it caught too many small market dips that do not represent genuine stress, making the signal noisy. The 10% threshold went the other way crashes that severe are too rare in our training window to give the model enough examples to learn from. The 8% threshold sits in the middle and gives the best balance between signal quality and sample size, confirming our original choice was the right one.

5.2 Alternative Rolling Windows

We tested alternative window lengths of 10 and 30 trading days against our 60-day baseline for the logistic regression models. The 30-day window performs slightly better, with the baseline logit achieving an AUC of 0.7183 versus 0.7009 for the 60-day baseline. However, we retain 60 days as the main specification for the logistic models since it was selected prior to the robustness checks on economic grounds. For the XGBoost specifications, a 20-day window was used to support the richer rate-of-change feature set, as noted in Section 3.2. The key finding is that results are not sensitive to window length — the model performs well across all specifications tested.

5.3 Sub-Period Analysis

To check whether the model holds up across different time periods we split the test period into two windows: January 2011 to December 2019 and January 2020 to December 2024. Results are reported for all three XGBoost specifications in Table 4.

Table 4: Sub-Period AUC Results

Period	V1	V1+Rotation	V2
2011–2019 pre-COVID	0.7057	0.7308	0.5893
2020–2024 post-COVID	0.6660	0.6911	0.5685

V1 and V1+Rotation hold up well across both periods — V1+Rotation is the strongest specification in both windows, confirming that the rotation signal adds consistent independent value rather than being concentrated in a specific period. The modest drop in post-COVID performance for both models is expected given that the 2020 to 2024 window includes the COVID shock, which as discussed in Section 5.4 is an exogenous event outside our model's scope. V2 struggles in both periods, confirming that the 40-day label does not improve discriminative performance despite its stronger backtest returns.

5.4 The COVID-19 Boundary Condition

COVID-19 was an external shock, not a buildup of internal market stress. The market was not fragile going into February 2020. It was ambushed from outside by a pandemic that no market-based signal could have predicted from correlation data alone. Our framework is designed to detect endogenous fragility not exogenous shocks like pandemics or geopolitical events. We therefore treat COVID-19 as a stated boundary condition of our model rather than a failure, consistent with Brunnermeier and Oehmke (2013).

6. Conclusion

This paper develops a network-based early warning system for US equity market crashes, treating cross-sector correlation structure as a set of daily vital signs that signal fragility before prices fall. All models are trained on 2000 to 2010 and tested strictly out-of-sample in 2011 to 2024.

Four results stand out. The baseline logistic regression outperforms the VIX-only benchmark (0.6693 vs 0.6449 AUC), confirming network biomarkers add predictive value beyond conventional risk measures. XGBoost V1 improves further to 0.7009. XGBoost V1+Rotation achieves the strongest discriminative result at 0.7264 AUC, confirming the rotation signal adds genuinely independent predictive value on top of network structure. A second-generation model (XGBoost V2) incorporates the rotation signal with an extended 40-day horizon, achieving a lower AUC of 0.6023 but generating warning signals two to four weeks earlier than V1. A V1-based long/cash strategy delivers a Sharpe ratio of 0.72 versus 0.68 for SPY with maximum drawdown reduced from 33.93% to 26.80%, while a V2-based strategy improves risk-adjusted performance further to a Sharpe of 0.90 at the cost of more time in cash. These findings partially resolve the accuracy-versus-timing trade-off in financial early warning systems. V1+Rotation achieves superior accuracy while preserving the 20-day horizon, narrowing but not fully closing the gap with V2's earlier warning capability. Key limitations include the model's boundary condition around exogenous shocks such as COVID-19, which market-based signals cannot anticipate from correlation data alone, and the conservative zero cash return assumption in the backtest which understates true strategy performance. Future

research should explore hybrid horizon architectures, alternative rotation metrics, and the incorporation of cross-asset signals such as credit spreads and yield curve data to further close the gap between prediction accuracy and trading utility.

7. Reference

Acemoglu, D., Ozdaglar, A., and Tahbaz-Salehi, A. (2015). Systemic risk and stability in financial networks. *American Economic Review*, 105(2), 564–608.

Adrian, T. and Brunnermeier, M. K. (2016). CoVaR. *American Economic Review*, 106(7), 1705–1741.

Baele, L., Bekaert, G., Inghelbrecht, K., and Wei, M. (2020). Flights to safety. *Review of Financial Studies*, 33(2), 689–746.

Billio, M., Getmansky, M., Lo, A. W., and Pelizzon, L. (2012). Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics*, 104(3), 535–559.

Brunnermeier, M. K. and Oehmke, M. (2013). Bubbles, financial crises, and systemic risk. *Handbook of the Economics of Finance*, 2, 1221–1288.

Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.

Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.

Longin, F. and Solnik, B. (2001). Extreme correlation of international equity markets. *Journal of Finance*, 56(2), 649–676.

Moskowitz, T. J. and Grinblatt, M. (1999). Do industries explain momentum? *Journal of Finance*, 54(4), 1249–1290.

Watts, D. J. and Strogatz, S. H. (1998). Collective dynamics of small-world networks. *Nature*, 393, 440–442.

Welch, I. and Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.